PRACTICAL-9

1.Aim

Interface Pi Camera with Raspberry Pi to capture image and video.

2. Prior Concepts

i) Basics of Raspberry Pi and Linux OS.

ii) Python programming fundamentals.

iii) GPIO and peripheral interfacing concepts.

3. New Concepts

i) Working of the Pi Camera Module.

ii) CSI (Camera Serial Interface) communication.

iii) Capturing images using Python.

iv) Recording videos using Python.

v) Using the picamera2 library.

vi) Camera preview and media storage handling.

4. Hardware Required

i) Raspberry Pi Camera Module

ii) Few RFID Cards or RFID Tags

iii)Jumper Wires

5. Procedure

Connect the finger print module to any one of the USB port of Raspi

i)Install Python3 on Raspi.

Write the below code in command prompt:

```
sudo apt install idle3
sudo apt install python3-picamera2 -y
sudo apt install libcamera-apps -y
pip install opencv-python
pip install numpy
```

6. Implementation with Python Code

Code:

```
from picamera2 import Picamera2
from time import sleep

#Initialize camera
pícam2 = Picamera2()

#Start camera
picam2.start()
```

```
# Allow camera to warm up
sleep(2)
#-----------------------------
#Capture Image
#-----------------------------
picam2.capture_file("captured_image.jpg")
print("Image Captured Successfully")

#---------------------------
# Record Video
#---------------------------
print("Video Recording Started")

#Start video recording
picam2.start_recording("captured_video.h264")

# Record for 10 seconds
sleep(10)

#Stop recording
picam2.stop_recording()

print("Video Recorded Successfully")

#Stop camera
picam2.stop()

print("Program Completed")
```

7. Results: